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Status	Finished
Started	Wednesday, 22 October 2025, 2:56 PM
Completed	Wednesday, 22 October 2025, 3:17 PM
Duration	21 mins 16 secs
Marks	50.00/50.00
Grade	5.00 out of 5.00 (100%)

Question 1

Correct

Mark 1.00 out of 1.00

What does "immutable infrastructure" help guarantee?

- ☐ a. Unpredictable configuration changes across environments that occur without centralized control or tracking mechanisms.
- ☒ b. Consistency between environments and reproducibility of deployments. ✓
- ☐ c. Regular manual interventions to update infrastructure components, increasing the risk of inconsistency and human error.
- ☐ d. Direct modifications made to production systems without automation or version control, leading to fragile and unreliable deployments.



The correct answer is: Consistency between environments and reproducibility of deployments.

Question 2

Correct

Mark 1.00 out of 1.00

What drives faster recovery from incidents?

- ☐ a. Delete monitoring dashboards to reduce noise and alerts.
- ☐ b. Restart systems randomly without understanding root causes.
- ☒ c. Automated rollback, playbooks, and practiced runbooks. ✓
- ☐ d. Rely on manual guesswork and ad-hoc troubleshooting.

The correct answer is: Automated rollback, playbooks, and practiced runbooks.

Question 3

Correct

Mark 1.00 out of 1.00



Which practice reduces accidental production changes?

- ☐ a. Avoid maintaining audit logs to simplify operations.
- ☒ b. Centralize deployments through the pipeline and remove direct access. ✓
- ☐ c. Share deployment credentials widely across teams without restrictions.
- ☐ d. Grant unrestricted SSH access to all production servers.

The correct answer is: Centralize deployments through the pipeline and remove direct access.

Question 4

Correct

Mark 1.00 out of 1.00

What reduces time to detect regressions in production?

- ☒ a. Rich observability, alerting, and canary analyses. ✓
- ☐ b. Extend release cycles to reduce deployment frequency and risk.
- ☐ c. Disable monitoring tools to simplify the production environment.
- ☐ d. Rely only on offline tests without real-time monitoring or feedback.



The correct answer is: Rich observability, alerting, and canary analyses.

Question 5

Correct

Mark 1.00 out of 1.00

Which practice reduces incident recurrence?

- ☐ a. Ignore incidents and avoid documenting root causes or resolutions.
- ☐ b. Promote a culture of blame and silence to discourage transparency.
- ☒ c. Blameless postmortems and follow-up fixes tracked through pipeline. ✓
- ☐ d. Focus only on reactive firefighting without long-term improvements.

The correct answer is: Blameless postmortems and follow-up fixes tracked through pipeline.

Question 6

Correct

Mark 1.00 out of 1.00

Why treat production as the primary test for real user behavior?

- ☐ a. Avoid testing in preproduction environments to save time.
- ☒ b. Real users reveal issues that preprod may miss; combine with safe progressive techniques. ✓
- ☐ c. Use manual exploration exclusively to identify user issues.
- ☐ d. Skip testing in other environments and rely solely on production feedback.

The correct answer is: Real users reveal issues that preprod may miss; combine with safe progressive techniques.

Question 7

Correct

Mark 1.00 out of 1.00

Which deployment approach helps when schema changes are needed?

- ☒ a. Use backward-compatible changes, toggles, and phased migrations. ✓
- ☐ b. Apply breaking schema changes directly to the live environment without safeguards or transitional strategies.
- ☐ c. Halt all deployment activities until the database is completely cleared of existing records and dependencies.
- ☐ d. Perform an immediate database drop to clear existing data and start fresh with the updated schema structure.



The correct answer is: Use backward-compatible changes, toggles, and phased migrations.

Question 8

Correct

Mark 1.00 out of 1.00

What is the role of IaC testing?

- ☐ a. Rely exclusively on manual inspection to verify infrastructure changes without automated validation processes.
- ☐ b. Bypass peer review practices and skip collaborative evaluation of infrastructure code before deployment.
- ☐ c. Consider infrastructure testing unnecessary and omit it from the development and deployment lifecycle.
- ☒ d. Validate infra changes and detect misconfigurations before promotion. ✓

The correct answer is: Validate infra changes and detect misconfigurations before promotion.

Question 9

Correct

Mark 1.00 out of 1.00

How should emergency patches be committed?

- ☒ a. Through an expedited, auditable pipeline path with post-hoc review. ✓
- ☐ b. Distribute emergency fixes via email attachments without version control.
- ☐ c. Skip audit trails to prioritize speed over traceability and accountability.
- ☐ d. Apply direct edits to production systems without documentation or review.

The correct answer is: Through an expedited, auditable pipeline path with post-hoc review.

Question 10

Correct

Mark 1.00 out of 1.00

How should pipeline failures be handled?

- ☐ a. Erase all failure logs to avoid clutter and skip any postmortem analysis.
- ☐ b. Continuously retry failed steps without investigation or corrective action.
- ☒ c. Fail fast, notify owners, capture diagnostics, and prevent unsafe promotions. ✓
- ☐ d. Conceal pipeline failures manually to prevent visibility and delay resolution.



The correct answer is: Fail fast, notify owners, capture diagnostics, and prevent unsafe promotions.

Question 11

Correct

Mark 1.00 out of 1.00

What is the consequence of not signing artifacts?

- ☐ a. Improves traceability by removing signature requirements and simplifying metadata.
- ☐ b. Strengthens audit trails through enhanced traceability and verification.
- ☒ c. Increased risk of tampering and weak provenance. ✓
- ☐ d. Enables easier rollback procedures by skipping verification steps.

The correct answer is: Increased risk of tampering and weak provenance.

Question 12

Correct

Mark 1.00 out of 1.00

What validates that the pipeline enforces policy?

- ☐ a. Trust the pipeline implicitly without any formal checks or reviews.
- ☐ b. Perform manual inspections only when incidents occur.
- ☐ c. Skip verification and assume pipeline behavior is correct.
- ☒ d. Periodic audits, tests of the pipeline, and verification of artifact signing and promotion paths. ✓

The correct answer is: Periodic audits, tests of the pipeline, and verification of artifact signing and promotion paths.

Question 13

Correct

Mark 1.00 out of 1.00

What should be included in a deployment artifact manifest?

- ☐ a. Random numbers generated during build without traceability.
- ☒ b. Checksums, version, source commit, and build environment info. ✓
- ☐ c. Editor configuration settings used during local development.
- ☐ d. Developer notes and personal annotations unrelated to deployment.



The correct answer is: Checksums, version, source commit, and build environment info.

Question 14

Correct

Mark 1.00 out of 1.00

What does Continuous Delivery emphasize about releases?

- ☐ a. Systems should minimize visibility into deployment processes and operational metrics, limiting the ability to monitor and diagnose issues effectively.
- ☐ b. Releases should be infrequent and treated as significant milestones, often requiring extensive planning and coordination.
- ☒ c. Deployments should be routine and automated so release on demand is possible. ✓
- ☐ d. Every release should involve manual steps and human oversight to ensure accuracy and control throughout the deployment process.

The correct answer is: Deployments should be routine and automated so release on demand is possible.

Question 15

Correct

Mark 1.00 out of 1.00

Which design reduces coupling between deploy and schema change?

- ☐ a. Make manual runtime edits to the database schema during deployment.
- ☐ b. Introduce immediate breaking changes without transitional safeguards.
- ☒ c. Multi-version data compatibility and feature toggles. ✓
- ☐ d. Apply monolithic, irreversible database changes that require downtime.

The correct answer is: Multi-version data compatibility and feature toggles.

Question 16

Correct

Mark 1.00 out of 1.00

Which telemetry helps detect user-impacting regressions early?

- ☐ a. Measure local machine time to estimate performance across environments.
- ☐ b. Use developer IDE plugins to monitor coding patterns and editor usage.
- ☒ c. Business KPIs, error rates, and latency percentiles. ✓
- ☐ d. Track the number of files changed in each commit as a regression indicator.



The correct answer is: Business KPIs, error rates, and latency percentiles.

Question 17

Correct

Mark 1.00 out of 1.00

What is a deployment "smoke" check for?

- ☐ a. Run long performance test suites to measure system load after deployment.
- ☐ b. Focus exclusively on database migration validation without user flow testing.
- ☒ c. Quick verification that core user flows are functional after deployment. ✓
- ☐ d. Replace observability tools with manual inspection of deployment outcomes.

The correct answer is: Quick verification that core user flows are functional after deployment.

Question 18

Correct

Mark 1.00 out of 1.00

Which anti-pattern increases configuration errors?

- ☐ a. Require peer-reviewed infrastructure changes before deployment.
- ☒ b. Allowing manual undocumented tweaks on running systems. ✓
- ☐ c. Apply versioned IaC with formal review and approval processes.
- ☐ d. Use automated drift detection to maintain environment consistency.

The correct answer is: Allowing manual undocumented tweaks on running systems.

Question 19

Correct

Mark 1.00 out of 1.00

Which practice eases compliance reporting?

- ☐ a. Use email approvals without storing formal records or audit trails.
- ☒ b. Keep auditable records: signed artifacts, promotion history, and pipeline logs. ✓
- ☐ c. Delete all logs after deployment to reduce storage and complexity.
- ☐ d. Avoid maintaining any logs or documentation related to deployments.



The correct answer is: Keep auditable records: signed artifacts, promotion history, and pipeline logs.

Question 20

Correct

Mark 1.00 out of 1.00

What is a benefit of canarying database-backed features?

- ☐ a. Replace schema design with runtime experimentation.
- ☐ b. Roll out features to all users immediately without monitoring.
- ☒ c. Expose feature for small user sets while observing DB-related effects before wider rollout. ✓
- ☐ d. Skip database testing and rely on application-level validation only.

The correct answer is: Expose feature for small user sets while observing DB-related effects before wider rollout.

Question 21

Correct

Mark 1.00 out of 1.00

How should experiments be cleaned up after dark launches?

- ☐ a. Leave stale flags in the codebase to accumulate over time.
- ☐ b. Keep flags hidden indefinitely without documentation or cleanup.
- ☒ c. Remove unused flags and associated code after rollout or retirement. ✓
- ☐ d. Never remove feature flags, even after rollout is complete.

The correct answer is: Remove unused flags and associated code after rollout or retirement.

Question 22

Correct

Mark 1.00 out of 1.00

Why automate environment promotion approvals where safe?

- ☐ a. Remove all human checks regardless of risk level or impact.
- ☒ b. To accelerate safe releases while reserving human review for high-risk changes. ✓
- ☐ c. Eliminate audit trails to streamline release processes.
- ☐ d. Introduce unpredictability into release schedules and outcomes.



The correct answer is: To accelerate safe releases while reserving human review for high-risk changes.

Question 23

Correct

Mark 1.00 out of 1.00

Which approach improves rollback predictability?

- ☐ a. Skip rollback planning and rely on ad-hoc fixes during incidents.
- ☒ b. Automated, tested revert procedures and versioned artifacts. ✓
- ☐ c. Apply manual fixes without documentation or testing.
- ☐ d. Restore backups manually during off-hours without automation.

The correct answer is: Automated, tested revert procedures and versioned artifacts.

Question 24

Correct

Mark 1.00 out of 1.00

How should deployment runbooks be maintained?

- ☐ a. Avoid creating any documentation related to deployment procedures.
- ☒ b. Keep them versioned, tested, and updated as systems change. ✓
- ☐ c. Leave runbooks outdated and never revise them after incidents.
- ☐ d. Store runbooks only in private chat channels without version control.

The correct answer is: Keep them versioned, tested, and updated as systems change.

Question 25

Correct

Mark 1.00 out of 1.00

Which test class is most valuable to run quickly on every commit?

- ☐ a. Conduct manual exploratory testing sessions to uncover edge cases and usability issues.
- ☐ b. Run full end-to-end test suites that simulate complete user journeys.
- ☒ c. Fast unit and integration tests that give quick feedback. ✓
- ☐ d. Execute long-running performance tests exclusively to measure system load.



The correct answer is: Fast unit and integration tests that give quick feedback.

Question 26

Correct

Mark 1.00 out of 1.00

Which artifact property is critical for traceability?

- ☐ a. Decorative background images used for personalization that do not contribute to software traceability or version tracking.
- ☐ b. Temporary files generated locally during development that are not preserved or tracked through version control systems.
- ☐ c. Configuration settings used by code editors to maintain formatting and style consistency across development environments.
- ☒ d. Embedded source commit and build identifier. ✓

The correct answer is: Embedded source commit and build identifier.

Question 27

Correct

Mark 1.00 out of 1.00

Which capability is essential for reproducible debugging?

- ☐ a. Avoid retaining traces to reduce storage and complexity.
- ☐ b. Rely solely on local logs without centralized traceability.
- ☐ c. Remove deployment metadata to simplify system architecture.
- ☒ d. Keep artifact ids, deploy timestamps, and correlated traces available. ✓

The correct answer is: Keep artifact ids, deploy timestamps, and correlated traces available.

Question 28

Correct

Mark 1.00 out of 1.00

Which gating criteria should pipelines include beyond build success?

- ☐ a. Evaluate only the size of pull requests without considering test or security outcomes.
- ☐ b. Assume build success alone is sufficient for safe deployment without further checks.
- ☐ c. Consider the age of the repository as the primary factor for promotion decisions.
- ☒ d. Test pass, security scan, and observability signals in target environment. ✓



The correct answer is: Test pass, security scan, and observability signals in target environment.

Question 29

Correct

Mark 1.00 out of 1.00

What is the main challenge of database migrations during continuous delivery?

- ☐ a. Always drop indexes during deployment regardless of performance impact.
- ☐ b. Assume schema changes are never needed and skip migration planning.
- ☒ c. Ensuring compatibility across versions while enabling fast rollbacks. ✓
- ☐ d. Run excessive numbers of tests without focusing on migration safety.

The correct answer is: Ensuring compatibility across versions while enabling fast rollbacks.

Question 30

Correct

Mark 1.00 out of 1.00

What does "promotion" mean in CD pipelines?

- ☐ a. Delete artifacts after initial testing to prevent further deployment.
- ☐ b. Restart servers manually to simulate environment transitions.
- ☐ c. Rename branches to reflect deployment stages without changing environments.
- ☒ d. Moving a validated artifact from one environment to the next (e.g., staging to prod). ✓

The correct answer is: Moving a validated artifact from one environment to the next (e.g., staging to prod).

Question 31

Correct

Mark 1.00 out of 1.00

Which role is responsible for defining pipeline policies?

- ☐ a. Leave policy definition unassigned and unmanaged across teams.
- ☐ b. Assign pipeline governance to a single operations person without collaboration.
- ☒ c. Cross-functional team including architects, SRE, security, and product stakeholders. ✓
- ☐ d. Rely solely on external consultants without internal ownership.



The correct answer is: Cross-functional team including architects, SRE, security, and product stakeholders.

Question 32

Correct

Mark 1.00 out of 1.00

How can teams reduce deployment anxiety?

- ☐ a. Avoid practicing rollbacks and rely on reactive fixes.
- ☒ b. Invest in automation, small changes, good observability, and rehearsed rollbacks. ✓
- ☐ c. Increase the size and complexity of each release to reduce frequency.
- ☐ d. Add more manual steps to the deployment process for control.

The correct answer is: Invest in automation, small changes, good observability, and rehearsed rollbacks.

Question 33

Correct

Mark 1.00 out of 1.00

What is a common CI anti-pattern?

- ☒ a. Long-lived integration branches with infrequent merges. ✓
- ☐ b. Making frequent, small code commits that are continuously integrated and tested to ensure rapid feedback and stability.
- ☐ c. Running automated test suites on pull requests to validate changes before merging into the main codebase.
- ☐ d. Regularly merging feature branches into the main branch to maintain alignment and reduce integration issues.

The correct answer is: Long-lived integration branches with infrequent merges.

Question 34

Correct

Mark 1.00 out of 1.00

How do you measure pipeline effectiveness?

- ☐ a. Measure how many manual steps are involved in each deployment.
- ☐ b. Count the number of lines of code in the repository to assess productivity.
- ☒ c. Lead time for changes, deployment frequency, MTTR, and change failure rate. ✓
- ☐ d. Track the duration of meetings to estimate team efficiency.

The correct answer is: Lead time for changes, deployment frequency, MTTR, and change failure rate.

Question 35

Correct

Mark 1.00 out of 1.00

Why are automated canary analyses important?

- ☐ a. They eliminate logging mechanisms from deployment pipelines to reduce data collection and storage overhead.
- ☐ b. They intentionally delay deployment rollouts to minimize the pace of software delivery and change adoption.
- ☒ c. They detect regressions early using statistical comparisons of metrics. ✓
- ☐ d. They serve as a substitute for continuous integration by handling build and test processes independently.

The correct answer is: They detect regressions early using statistical comparisons of metrics.

Question 36

Correct

Mark 1.00 out of 1.00

What is the role of automated security checks in CD?

- ☐ a. Rely solely on manual security reviews without automation or integration into pipelines.
- ☐ b. Skip all security scans to streamline the deployment process and reduce overhead.
- ☒ c. Early detection of vulnerabilities and blocking unsafe artifacts from promotion. ✓
- ☐ d. Perform vulnerability scans only once a month regardless of release frequency.

The correct answer is: Early detection of vulnerabilities and blocking unsafe artifacts from promotion.

Question 37

Correct

Mark 1.00 out of 1.00

What is the effect of insufficient observability retention?

- ☒ a. Harder to investigate past incidents and correlate regressions. ✓
- ☐ b. Have no impact on incident investigation or system reliability.
- ☐ c. Improve query speed by discarding historical data regularly.
- ☐ d. Simplify long-term debugging by removing trace data.



The correct answer is: Harder to investigate past incidents and correlate regressions.

Question 38

Correct

Mark 1.00 out of 1.00

What does signed release metadata provide?

- ☐ a. Offer no additional value beyond basic versioning.
- ☐ b. Display visual labels without verifying origin or integrity.
- ☒ c. Cryptographic proof of artifact origin and integrity. ✓
- ☐ d. Generate random numbers to identify artifacts without traceability.

The correct answer is: Cryptographic proof of artifact origin and integrity.

Question 39

Correct

Mark 1.00 out of 1.00

What is the role of synthetic tests in deployment?

- ☐ a. Replace real-user metrics with simulated performance data.
- ☐ b. Serve no purpose and should be excluded from deployment pipelines.
- ☐ c. Generate artificial load without validating user functionality.
- ☒ d. Provide predictable, repeatable checks to detect regressions in production-like traffic. ✓

The correct answer is: Provide predictable, repeatable checks to detect regressions in production-like traffic.

Question 40

Correct

Mark 1.00 out of 1.00

How should feature flag state be tested?

- ☒ a. Validate both on and off behavior in integration environments. ✓
- ☐ b. Validate feature flags only in production environments without prior testing.
- ☐ c. Assume feature flags always behave correctly and never require verification.
- ☐ d. Avoid testing feature flags entirely and rely on default behavior.



The correct answer is: Validate both on and off behavior in integration environments.

Question 41

Correct

Mark 1.00 out of 1.00

How should test environments be provisioned for parity?

- ☐ a. Leave environments outdated and inconsistent with production setups.
- ☐ b. Use developer machines exclusively without replicating production characteristics.
- ☒ c. Provisioned via IaC to match production characteristics where feasible. ✓
- ☐ d. Manually configure environments each time without automation or standardization.

The correct answer is: Provisioned via IaC to match production characteristics where feasible.

Question 42

Correct

Mark 1.00 out of 1.00

Why is a blameless postmortem culture important?

- ☐ a. Avoid documenting incidents to prevent scrutiny or accountability.
- ☐ b. Suppress discussion of failures to maintain team morale.
- ☐ c. Assign personal blame to individuals responsible for incidents.
- ☒ d. Encourage open investigation and continuous improvement without fear. ✓

The correct answer is: Encourage open investigation and continuous improvement without fear.

Question 43

Correct

Mark 1.00 out of 1.00

Which is a sign of healthy progressive delivery?

- ☐ a. Require extensive manual intervention for every deployment.
- ☒ b. Frequent small releases with low impact and fast recovery. ✓
- ☐ c. Experience frequent large-scale incidents after each release.
- ☐ d. Release software infrequently and only in large batches.



The correct answer is: Frequent small releases with low impact and fast recovery.

Question 44

Correct

Mark 1.00 out of 1.00

How should IaC drift be remediated?

- ☐ a. Ignore configuration drift and allow environments to diverge indefinitely.
- ☒ b. Detect via drift checks and remediate by reapplying versioned IaC. ✓
- ☐ c. Apply manual, undocumented updates to infrastructure components.
- ☐ d. Make random edits directly on servers without tracking changes.

The correct answer is: Detect via drift checks and remediate by reapplying versioned IaC.

Question 45

Correct

Mark 1.00 out of 1.00

What is the impact of long CI queues on developer workflow?

- ☐ a. Reduce the need for testing by batching changes together.
- ☐ b. Improve merge reliability by delaying integration steps.
- ☐ c. Accelerate feedback cycles by skipping validation steps.
- ☒ d. Slower feedback loops and increased context switching. ✓

The correct answer is: Slower feedback loops and increased context switching.

Question 46

Correct

Mark 1.00 out of 1.00

What is an appropriate response to a canary anomaly that is ambiguous?

- ☐ a. Delete the canary data to avoid investigating potential issues.
- ☒ b. Pause and increase observation or reduce exposure while investigating. ✓
- ☐ c. Proceed with full rollout immediately without further validation.
- ☐ d. Disable monitoring tools to prevent further alerts or analysis.



The correct answer is: Pause and increase observation or reduce exposure while investigating.

Question 47

Correct

Mark 1.00 out of 1.00

Why are runbooks important post-rollback?

- ☐ a. Consider runbooks unnecessary and rely solely on tribal knowledge for incident response.
- ☐ b. Increase confusion by introducing inconsistent or outdated recovery steps.
- ☐ c. Replace automated rollback mechanisms with manual recovery instructions.
- ☒ d. Provide step-by-step recovery guidance and context for on-call engineers. ✓

The correct answer is: Provide step-by-step recovery guidance and context for on-call engineers.

Question 48

Correct

Mark 1.00 out of 1.00

Which method identifies subtle performance regressions early?

- ☐ a. Rely only on unit tests to measure system performance.
- ☒ b. Continuous performance benchmarking and monitoring percentiles. ✓
- ☐ c. Conduct ad-hoc load tests once a year without consistent benchmarks.
- ☐ d. Ignore performance metrics and focus only on functional correctness.

The correct answer is: Continuous performance benchmarking and monitoring percentiles.

Question 49

Correct

Mark 1.00 out of 1.00

Which behavior supports a culture for effective DevOps?

- ☐ a. Avoid team retrospectives to prevent open discussion of failures or improvements.
- ☐ b. Assign responsibilities to isolated individuals without shared accountability.
- ☐ c. Promote blame-focused reviews and maintain strict separation between teams.
- ☒ d. Shared ownership, blameless postmortems, and cross-functional collaboration. ✓



The correct answer is: Shared ownership, blameless postmortems, and cross-functional collaboration.

Question 50

Correct

Mark 1.00 out of 1.00

How often should pipeline infrastructure be updated?

- ☐ a. Apply random, untested changes without planning or review.
- ☒ b. Regularly with automation and testing to stay current and secure. ✓
- ☐ c. Perform updates only after major incidents or outages.
- ☐ d. Never update infrastructure to avoid introducing changes.

The correct answer is: Regularly with automation and testing to stay current and secure.

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